EVERY SEQUENCE OF HOMOMORPHISMS IS DOMINATED BY A SEQUENCE OF ADMISSIBLE MONOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two-term workhorse is Lemma lemma-make-injective (in-source, navigation stub); the induction step applies it to the composition M_{n-1} -> L_{n-1} -> L_n, which preserves the homotopy-equivalence property by functoriality of homotopies.

Editorial note. The label lemma-sequence-maps-split names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Two-term case. Same unit, Lemma lemma-make-injective. The base case n = 2 of the induction is exactly Lemma lemma-make-injective, and the induction step applies that lemma to a composition.

Source reference. Original reference 013N: sources/stacks-dga/pinned-original-source.tex, source lines 777–777. Prerequisite pointer invoked at proof line(s) 830; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let $L_1 \to L_2 \to \ldots \to L_n$ be a sequence of composable homomorphisms of differential graded A-modules. There exists a commutative diagram*

$$\begin{array}{ccccccc} L_1 & \longrightarrow & L_2 & \longrightarrow & \ldots & \longrightarrow & L_n \\ \uparrow & & \uparrow & & & & \uparrow \\ M_1 & \longrightarrow & M_2 & \longrightarrow & \ldots & \longrightarrow & M_n \end{array}$$

in $Mod_{(A,d)}$ such that each $M_i \to M_{i+1}$ is an admissible monomorphism and each $M_i \to L_i$ is a homotopy equivalence.

Proof. The case $n = 1$ is without content. Lemma 013N is the case $n = 2$. Suppose we have constructed the diagram except for M_n. Apply Lemma 013N to the composition $M_{n-1} \to L_{n-1} \to L_n$. The result is a factorization $M_{n-1} \to M_n \to L_n$ as desired. □

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